

1. **Experiment:** Construction of the dCas9-AID*Δ System
2. **Time:** 2026.03.18-2026.04.02
3. **Member:** Kaiqing Zhang, Yang Jin, Binxuan Zhang, Qiwen Jiang, Zhangdong Liu, Xinru Yuan
4. **Method:**

(1) Plasmid Maps and dCas9-AID*Δ Sequence

- ① According to the plasmid maps below, the dCas9-AID*Δ lentiviral system was constructed using the transfer plasmid pCDH-dCas9-AID*Δ, the envelope plasmid pMD2.G, and the packaging plasmid pCMVR8.74. The plasmid maps shown here are schematic representations for documenting the constructs used in this experiment.

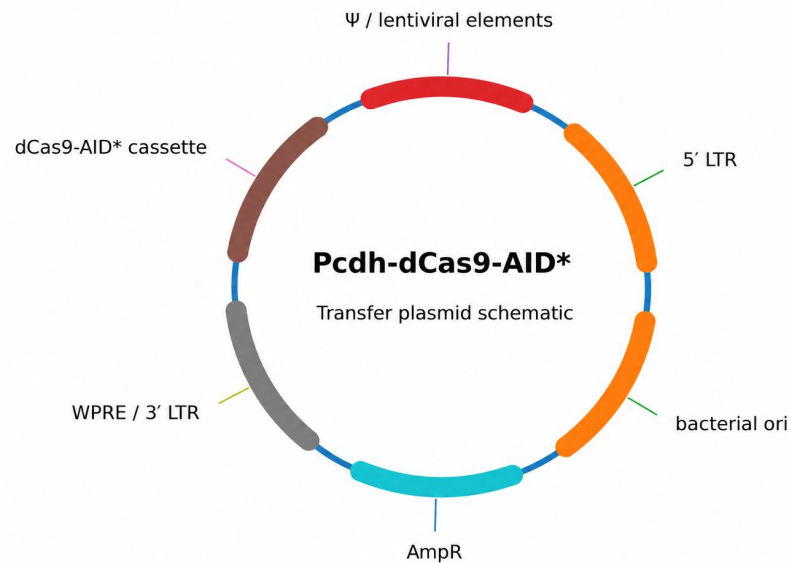
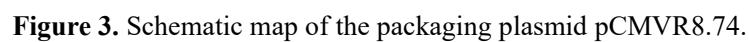
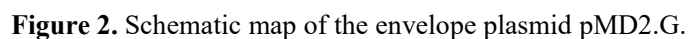


Figure 1. Schematic map of the transfer plasmid pCDH-dCas9-AID*Δ.



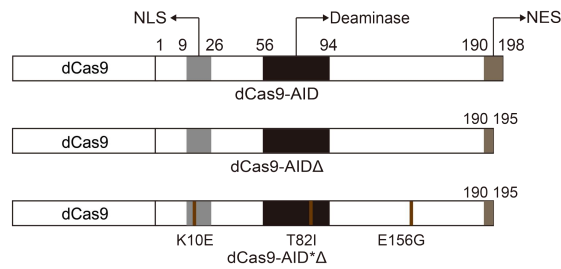


Figure 4. Schematic of dCas9-AID variants and the final expression cassette.

Domain organization of dCas9-AID, dCas9-AIDΔ, and dCas9-AID*Δ. The NLS (aa 9-26), deaminase domain (aa 56-94), and NES (aa 190-198) are indicated. Compared with dCas9-AID, dCas9-AIDΔ lacks the C-terminal three amino acids. dCas9-AID*Δ is based on dCas9-AID and additionally carries the K10E, T82I, and E156G mutations. The NLS, deaminase domain, and NES are retained in these constructs.

② dCas9-AID*Δ amino acid sequence:

CD8-SP-FLAG tag-GGGGS-GGGGS-CD8α TM-T2A-dCas9-nuclear localization sequence-AID*

CD8-SP: MALPVTALLPLALLHAARP

FLAG tag: DYKDDDDK

GGGGS-GGGGS

CD8α TM: IYIWAPLAGTCGVLLSLVITLY

T2A: EGRGSLTCTGDVEENPGP

dCas9:

MDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNTDRHSIKKNLIGALLFDSGETAEAT
RLKRTARRRYTRRKNRICYLQEIFSNEMAKVDDSFHRLVESFLVEEDKKHERHPIFGNIVDE
VAYHEKYPTIYHLRKKLVDSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQL
VQTYNQLFEEENPINASGVDAKAILSARLSKSRRLENLIAQLPGEKKNGLFGNLIASLGLTPN
FKSNFDLAEDAQLSKDQYDDDLNLLAQIGDQYADLFLAAKNLSDAILSDILRVNTEIT
KAPLSASMIKRYDEHHQDLTLLKALVRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKF
IKPILEKMDGTEELLVKNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNRE
KIEKILTRIPYYVGPLARGNSRFAWMTRKSEETITPWNFEVVDKGASQSFIERMTNFDKN
LPNEKVLPHKSLLEYFTVYNELTKVKYVTEGMRKPAFLSGEQKKAIVDLLFKTNRKVTVK
QLKEDYFKKIECFDSVEISGVEDRFNASLGTYHDLLKIIKDKDFLDNEEKEDILEDIVLTTLF
EDREMIEERLKYAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDQSGKTILDFLKSDG
FANRNFMQLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIAKKGILQTVKVVDELVK
VMGRHKPENIVIEMARENQTTQKGQKNSRERMKRIIEGKELGSQILKEHPVENTQLQNEKL
YLYYLQNGRDMYVDQELDINRLSDYDVAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNVPS
EEVVKKMKNYWRQLLNAKLITQRKFDNLTKAERGGSELDAKAGFIKRQLVETRQITKHVAQ
ILDSRMNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKVREINNYHHAHDAYLNAVVG
TALIKKYPKLESEFVYGDYKVYDVRKMIKSEGEIGKATKAYFFYSNIMNFFKTEITLANGEI
RRRPLIETNGETGEIVWDKGRDFATVRKVLSMPQVNIVKKTEVQTGGFSKESILPKRNSDKLI
ARKKDWDPKKYGGFDSPTVAYSVLVAKVEKGKSKKLKSVKELLGITIMERSSEKPNIDFL
EAKGYKEVKKDLIKLPKYSLEFENGRKRMLASAGELQKGNELALPSKYVNFYLYLASHYE
KLKGSPEDEQKQLFVEQHKHYLDEIEQISEFSKRVLADANLDKVL SAYNKHDKPIREQA
ENIIHLFTLTNLGAPAAFKYFDTTIDRKRYTSTKEVLDTLIHQSIETGLYETRIDLSQLGGD

Nuclear localization sequence:

EGAPKKKRKVGSSGSPKKKRKVGSDALDDFDLDMLGSDALDDFGGGS

AID*Δ (already have three point mutations):

MDSLLMNRREFLYQFKNVRWAKGRRETYLCYVVKRRDSATSFSLDFGYLRNKNKGCHVELL
FLRYISDWDLDPGRCYRVTWFWISWSPCYDCARHVADFLRGPNLSLRIFTARLYFCEDRKAE
PEGLRRLHRAGVQIAIMTFKDYFYCWNTFVENHGRTFKAWEGLHENSVRLSRQLRRILLP

① Notes:

Point mutation 1

Residue 10: Lysine (K) → Glutamic acid (E)

Point mutation 2

Residue 82: Threonine (T) → Isoleucine (I)

Point mutation 3

Residue 156: Glutamic acid (E) → Glycine (G)

C-terminal truncation

Deletion of the last three amino acids: removal of the nuclear export signal (NES)

(2) Lentivirus Packaging and Induced Culture

- ② One day before transfection, Hek293T cells was passaged and evenly plated on 100 mm culture dishes (at $8-10 \times 10^6$ cells per dish). Cells were cultured at 37 °C in a 5% CO₂ incubator. The medium was replaced with 10.5 mL of pre-warmed, antibiotic-free medium (5% FBS) 2-4 h before transfection.
- ③ A mixture of three plasmids (transfer plasmid + envelope plasmid + packaging plasmid) in equimolar amounts was prepared, and transfection was performed using the PEI transfection method with the Hieff Trans@Polyethylenimine Linear. The transfection mixture composition per well was as follows:

Table 1. Transfection mixture composition.

Transfection tube	Reagent	Volume/Amount
A	PEI (1 mg/mL)	90 µL
	Antibiotic-free medium	910 µL (to 1 mL total)
	Transfer plasmid (pCDH-dCas9-AID*Δ)	15 µg
	Envelope plasmid (pMD2G)	5 µg
B	Packaging plasmid (pCMVR8.74)	10 µg
	Antibiotic-free medium	Up to 1mL

- ④ The PEI solution (Tube A) and the plasmid mixture (Tube B) were gently mixed separately, then combined by adding Tube B dropwise into Tube A using a pipette gun. The mixture was vortexed briefly and incubated at room temperature for 15 min to allow complex formation. The transfection complexes were then added dropwise to the Hek293T cells and the plates were returned to the incubator.
- ⑤ Viral supernatant collection: At 48 h and 72 h post-transfection, collect the viral supernatant from Hek293T cells, filter through a 0.45µm filter, and use immediately or store at -80 °C .
- ⑥ Cell Inoculation
- Collect Raji cell from a culture flask.
 - Centrifuge at 300×g for 5 min at room temperature.
 - Carefully remove the supernatant and resuspend the cell pellet in an appropriate volume of complete medium (RPMI-1640 with 10% FBS and 1% penicillin-streptomycin).
 - The cell density was determined using an automated cell counter, and the suspension was diluted to a final concentration of 1×10^6 cells/mL.
 - Add the calculated volume of cells to an appropriate volume of complete medium to achieve a final

concentration of 1×10^6 cells/mL.

- For each well of a 6-well plate (or equivalent dish), 2 mL of this cell suspension (2×10^6 cells) was dispensed. If using a 10 cm dish, adjust the volume to 10 mL (1×10^7 cells). The plates were incubated at 37 °C in a 5% CO₂ incubator for 24 h prior to infection.
- ⑦ Infection: To each well, Polybrene was added to a final concentration of 8 µg/mL. The multiplicity of infection (MOI) was set to 0.3, using the volume calculated as $(0.3 \times \text{cell number}) / \text{virus titer}$, was added to the wells. The plates were then incubated at 37 °C for 12-16 h, then replace the viral inoculum with fresh complete medium..

(3) Single-Cell Sorting and Planting

Fluorescence-activated cell sorting (FACS) uses target cells' specific fluorescent signals to isolate individual cells into 96-well plates, ensuring clonal growth from a single cell. Second, in vitro culture mimics physiological conditions to sustain cell viability and proliferation, while inverted microscopy tracks clonal expansion. For sequencing, gentle centrifugation preserves cell integrity during collection; cell lysis releases nucleic acids, enabling high-quality sequencing of clonally pure single-cell samples. Live cells were stained with an anti-FLAG antibody (PE- or APC-conjugated) at 4 °C in the dark for 30 min, washed, and then sorted.

- ① FACS was performed under the following conditions: Nozzle size: 100µm. Sort mode: Single-cell deposition (1 drop/well). Gating strategy: FLAG-positive cells were gated based on PE or APC fluorescence intensity.
- ② Single cells were sorted directly into 96-well plates pre-filled with 150 µL Raji cell complete medium (RPMI 1640 + 10% FBS + 1% penicillin-streptomycin).
- ③ Negative control wells (medium only, no cells) were included in each plate.

(4) Culture and Monitoring:

- ① All sorted plates were transferred to cell incubator (37 °C, 5% CO₂) and cultured for 7-10 days to allow clonal expansion.
- ② Half-medium change was conducted on Day 3: 75 µL of spent medium was aspirated and replaced with 75 µL fresh pre-warmed Raji cell complete medium.
- ③ Clonal growth was monitored daily using an inverted microscope. Positive clones were sequentially transferred from 96-well plates to 24-well plates, 6-well plates, T25 flasks, and T75 flasks, with centrifugation and resuspension in fresh medium at each transfer, and finally expanded to 10^7 - 10^8 cells for subsequent experiments.